

Efficient Self-Verification: Understanding LaSeR for LLM Reasoning

Unifying Reasoning and Verification via Last-Token Self-Rewarding



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Outline

- **Introduction:** The Evolution of LLM Training (SFT → RLVR)
- **Problem Statement:** The Test-Time Verification Gap
- **Methodology:** LaSeR (Reinforcement Learning with Last-Token Self-Rewarding)
- **Theoretical Framework:** Deriving the Closed-Form Solution
- **Experimental Results:** Accuracy and Efficiency Gains
- **Critical Analysis:** Strengths and Limitations
- **Conclusion:** Towards Efficient System 2 Reasoning

Introduction:

The Evolution of Reasoning

- **Supervised Fine-Tuning (SFT):** The foundation of instruction following, but lacks deliberate reasoning incentives.
- **RLHF (Reinforcement Learning from Human Feedback):** Aligns models with human preference, yet suffers from subjective reward signals.
- **RLVR (RL with Verifiable Rewards) [3]:**
 - Current SOTA (e.g., DeepSeek-R1 [2], OpenAI o1) utilizes ground-truth outcomes (math/code) to incentivize "System 2" reasoning chains.
 - Mechanism: Rewards reasoning paths based on final answer consistency.
- **The Next Frontier:** How do we maintain this rigorous verification when ground truth is absent (i.e., at inference time)?

How does RLVR work?

- **System Message:** “Wrap the final answer in <ANSWER>...</ANSWER> tags.”
- **Question:** “What is $4 * 3$?”
- **Verifiable Answer:** “12”
- **Sampled Rollouts:**
 - *Answer 1:* “4 groups of 3 is equivalent to $4 + 4 + 4$, which equals 12.
<ANSWER>12</ANSWER>” - **CORRECT (Reward: 1)**
 - *Answer 2:* “The numbers are 4 and 3. Adding them together gives $4 + 3 = 7$.
<ANSWER>7</ANSWER>” - **INCORRECT (Reward: 0)**
 - *Answer 3:* “4 times 3 is a standard multiplication fact. The result is 14.
<ANSWER>14</ANSWER>” - **INCORRECT (Reward: 0)**
 - *Answer 4:* “If you combine the digits 4 and 3, you get the number 43.
<ANSWER>43</ANSWER>” - **INCORRECT (Reward: 0)**
- **Policy Update:** Update the LLM model weights to maximize the likelihood of the correct reasoning path (*Answer 1*) relative to the incorrect paths.

Problem Statement:

The Test-Time Blind Spot

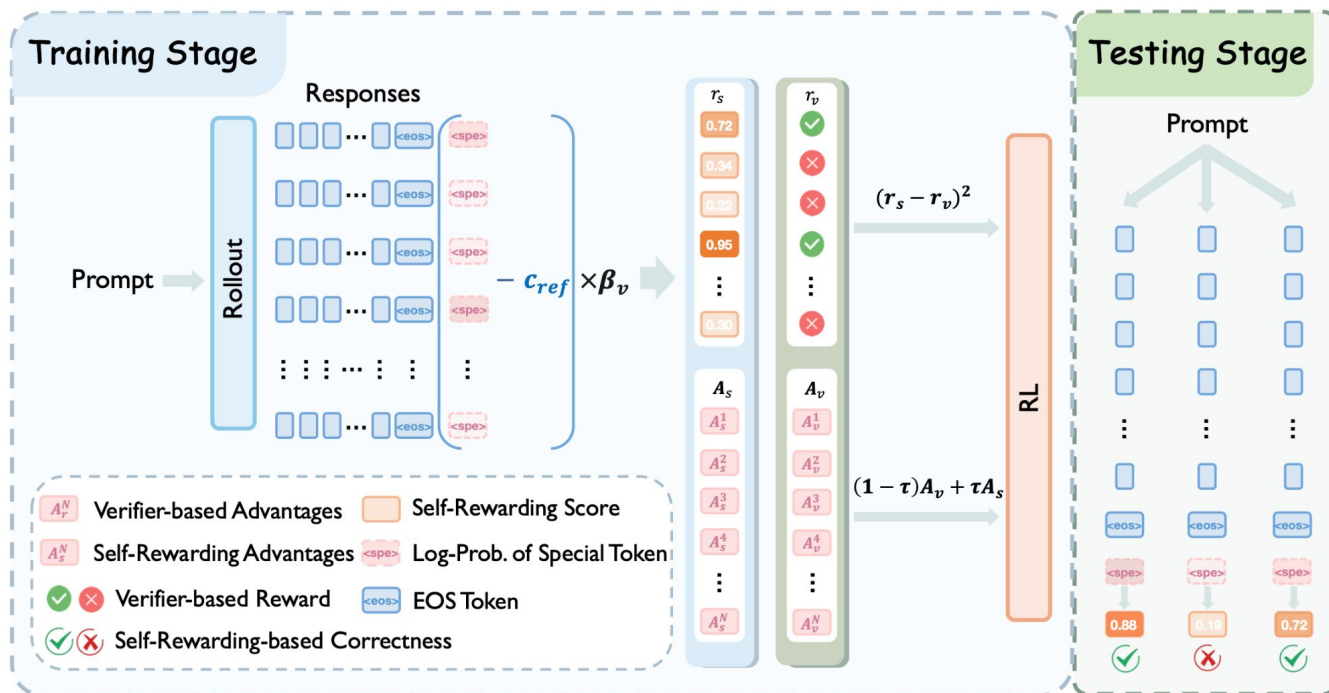
- **The Constraint:** Standard RLVR relies on ground-truth labels which are unavailable during deployment.
- **Current Inefficient Solutions:**
 - External Verifiers: Training separate Reward Models (ORMs/PRMs) is computationally expensive and parameter-heavy.
 - Sequential Self-Verification: Asking the model to "double-check" its answer requires a second generation pass, doubling inference costs.
- **Research Question:** Can we unify reasoning and verification within a single policy model at almost zero additional cost?

Methodology:

LaSeR Overview

- **Core Concept:** LaSeR (Reinforcement Learning with Last-Token Self-Rewarding).
- **Mechanism:**
 - The model predicts a specific "verification token" (e.g., <vision_start>) immediately after the solution.
 - The log-probability of this token serves as the self-rewarding score.
- **Implementation:**
 - Augments standard RLVR loss with a Mean Squared Error (MSE) loss.
 - Aligns the last-token probability with the true verifier reward during training.
- **Efficiency:** Requires only one additional token inference per generation.

LaSeR Methodology [1]



Theoretical Framework: The Derivation

- **RL Objective For Post-Training [4]:**

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi} [r(x, y)] - \beta \mathbb{D}_{KL}[\pi(y|x) || \pi_{ref}(y|x)]$$

- **Closed-Form Optimal Policy:**

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{ref}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right) \quad Z(x) = \sum_y \pi_{ref}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

- **Solve Equation For The Reward:** on the next slide!
(also conditioning on verification token is added)

Theoretical Framework: The Derivation

- **Closed-Form Solution [4]:**

$$r_v(x, y) = \beta_v \log \frac{\pi_\theta(z_c|x, y)}{\pi_{ref}(z_c|x, y)} + \beta_v \log Z(x, y)$$

$\Uparrow$

$$Z(x, y) = \sum_z \pi_{ref}(z|x, y) \exp\left(\frac{1}{\beta_v} \hat{r}(x, y, z)\right)$$

- Where z_c is the verification token and π_{ref} is the reference model.
- **The Simplification:**
 - For unused special tokens, $\pi_{ref}(z_c|x, y) \approx 0$ (negligible probability).
 - Therefore, the partition function $Z(x, y) \approx 1$ allowing the term $\log Z(x, y)$ to be discarded.
- **Result:** The reasoning reward is directly equivalent to the scaled log-probability ratio of the last token.

Practical Optimization Techniques

- **Reference Log-Probability as Constant:**
 - Empirical analysis shows $\pi_{ref}(z_c|x, y)$ is effectively constant (c_{ref}) across inputs.
 - **Benefit:** Eliminates the need for a forward pass through the reference model during self-rewarding
- **Self-Rewarding-based Advantages (SWA):**
 - Integrates the continuous self-rewarding score into the RL advantage estimation.
 - Formula: $(1 - \tau)A_v + \tau A_s$ (blends verifier and self-reward advantages, e.g. 10% self-reward signal and 90% external verifier signal).
 - **Benefit:** Provides dense signals to smooth out sparse binary rewards.
- **Loss Re-Weighting:** Balances correct/incorrect class distribution to prevent mode collapse.
- **Final Shortened Formula ($\beta_v = 0.1$):**

$$r_v(x, y) \approx \beta_v \log \pi_{\theta}(z_c|x, y) - \beta_v c_{ref}$$

Advantages: Why LaSeR?

- **Computational Efficiency:**
 - Replaces separate reward models (e.g., 72B parameters) with a single token check.
 - Avoids the 2x latency penalty of sequential self-verification.
- **Performance:**
 - Reasoning: Outperforms standard GRPO (e.g., Open-Reasoner-Zero-7B: **45.5% vs 45.0%** average accuracy).
 - Verification: Achieves **~78% F1 score** in self-verification, matching state-of-the-art external verifiers with a much higher parameter count.
- **Inference Scaling:** Enables effective weighted majority voting at test time without external tools.

Reasoning and Self-Verification [1]

Method	Reasoning Accuracy						Self-Verification F1 Score					
	MATH-500	AMC-23	AIME-24	AIME-25	Olym.-Bench	Avg.	MATH-500	AMC-23	AIME-24	AIME-25	Olym.-Bench	Avg.
<i>Open-Reasoner-Zero-7B</i>												
Base	81.9	60.3	15.6	15.1	46.9	44.0	26.7	51.3	45.9	55.2	37.5	43.3
GRPO	83.1	61.9	18.1	15.0	47.1	45.0	57.1	44.8	14.6	28.1	49.5	38.8
LaSeR	82.8	62.7	19.1	15.1	47.8	45.5	87.2	79.7	64.6	77.7	78.7	77.6
- SWA	83.2	62.6	19.0	14.5	47.6	45.4	87.5	77.7	63.3	77.3	77.9	76.7

External Reward Model Comp. [1]

Self-Verification F1 Score

Method	MATH500	AMC23	AIME24	AIME25	Olym.	Avg.
<i>Generator: Open-Reasoner-Zero-7B-LaSeR</i>						
Qwen2.5-Math-7B-PRM800K (7B RM)	56.3	42.5	51.4	50.8	38.5	47.9
Qwen2.5-Math-PRM-7B (7B RM)	86.0	79.6	70.8	67.3	76.0	75.9
Qwen2.5-Math-RM-72B (72B RM)	86.8	79.4	71.0	71.4	75.5	76.8
LaSeR (7B Self-Rewarding)	87.2	79.7	64.6	77.7	78.7	77.6

Experimental Results

- **Benchmarks:** MATH500, AMC23, AIME24, AIME25, OlympiadBench.
- **Used Base Model Architectures:** LLaMA-3.2 (3B) and Qwen-2.5 (7B).
- **Key Findings:**
 - LaSeR consistently yields higher Pass@1 accuracy than baselines.
 - Self-verification capability remains robust even on difficult datasets (e.g., AIME25) where reasoning accuracy is low.
 - "Token Hijacking" (using <vision_start>) successfully encodes semantic correctness without disrupting generation.

Critical Analysis & Limitations

- **Generalization Gap:**
 - Highly effective in deterministic domains (Math).
 - Performance degrades in general reasoning (e.g., MMLU-Pro) due to noisy training verifiers and subjective ground truth.
- **Prompt Sensitivity:** The mechanism is sensitive to the specific prompt templates used to trigger the verification state.
- **Aggregation Heuristic:** Reliance on the last token assumes the model can aggregate all prior reasoning uncertainty into a single step, which may be susceptible to length bias in complex chains.
- **Non-Standard Mechanism:** Additional token generation step after the “EOS” token is non-standard.

Conclusion & Outlook

- **Summary:** LaSeR theoretically and empirically proves that LLMs can act as their own verifiers by utilizing the probability distribution of the final token.
- **Impact:** Unifies generation and verification into a single, compute-optimal workflow.
- **Future Directions:**
 - Extending "Last-Token" verification to "Step-by-Step" (PRM) verification for finer granularity (e.g. could be further broken down to generating the self-reward signal at each token generation step).
 - Improving generalization to non-STEM domains.
 - Removing the dependence on the single additional forward pass after the "EOS" token.

References

- **[1]** Yang, W., et al. (2025). LaSeR: Reinforcement Learning with Last-Token Self-Rewarding. arXiv:2510.14943v1.
- **[2]** Guo, D., et al. (2025). DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning.
- **[3]** He, Z., et al. (2025). DeepMath-103K: A large-scale, challenging, decontaminated, and verifiable mathematical dataset.
- **[4]** Rafailov, R., et al. (2023). Direct preference optimization: Your language model is secretly a reward model.